

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

SSC PART-I (CLASS 9) PRACTICE EXAMINATION

BIOLOGY PAPER 18 | ANSWER KEY (Version A)

Syllabus: Chapters 1 to 7 (revision of the whole course studied so far)

Total Marks: 50

Section A: 10

Section B: 24

Section C: 16

Marking rules used on this paper.

- A missing **name** or a missing **unit** loses $\frac{1}{2}$ **mark**, the same in every question. It never cancels correct working or a correct equation.
- An explanation that stops before the concluding sentence the question asked for loses $\frac{1}{2}$ **mark** on that part.
- Where a question asks for a stated number of points, mark only that number. Extra correct points are not penalised.
- Q.4(ii) is a textbook exercise question for which the book prints no model answer. Credit **any** organelle named with a sound reason.
- **Nothing on this paper comes from Chapter 8, 9 or 10.** Every question sits inside Chapters 1 to 7.

SECTION A: ANSWER KEY (10 × 1 = 10 MARKS)

1. Inserting the human insulin gene into bacteria: **(b) biotechnology** 1

Biotechnology is the manipulation of living organisms, or their components, to make useful products. Genetically engineering bacteria so that they make human insulin is the book's own example. Anatomy is internal structure, physiology is the working of organs, and pharmacy is not the branch that does the gene transfer.

Source: Ch 1, section 1.3 and Table 1.2 (Biotechnology), and Exercise Section I, MCQ 6, book p17. SLO 5 and SLO 6. **NEW**

2. Colonial organization is the unique feature of: **(c) Protista** 1

The book describes Kingdom Protista as eukaryotic organisms that are **unicellular or colonial**. Animalia and Plantae are multicellular, and Fungi are mostly multicellular with a chitin cell wall.

Source: Ch 2, section 2.4 (Four kingdoms of Domain Eukarya), and Exercise Section I, MCQ 12, book p29. SLO 4. **NEW**

3. Correct statement about ribosomes: **(a) made in the nucleus, then work in the cytoplasm, where they make proteins** 1

The subunits of the ribosome are assembled in the nucleolus inside the nucleus, and the ribosome then works in the cytoplasm. (b) is wrong because ribosomes are found in prokaryotic cells too. (c) is wrong because ribosomes are **not** bound by any membrane. (d) is wrong because a large number of ribosomes are scattered free in the cytoplasm.

Source: Ch 3, Ribosomes (book p34) and chapter Summary point 3 on the nucleolus (book p43), and Exercise Section I, MCQ 11, book p46. SLO 3. **NEW**

4. What halves the chromosome number in meiosis: **(d) separation of homologous chromosomes during meiosis I** 1

Meiosis I is the **reduction division**. In anaphase I the two members of each homologous pair go to opposite poles, so each pole gets a haploid set. DNA replication in interphase I doubles the DNA instead of halving it, crossing over only exchanges pieces, and the separation of sister chromatids in meiosis II keeps the number the same.

Source: Ch 4, section 4.5 (Meiosis I, Anaphase I), book p55, and Exercise Section I, MCQ 10, book p61. SLO 2 and SLO 3. **RE-TEST** of the mitosis versus meiosis mix-up in exam-17 Q.3(ii)(c).

5. Removes pathogens, fights infections and helps in healing: **(b) immune system** 1

Table 5.2 gives the immune system the homeostatic role "by removing pathogens, fighting infections and helping in healing". The lymphatic system also defends the body, but its printed role in the same table is "maintain tissue fluid homeostasis". That is the pair she must separate.

Source: Ch 5, Table 5.2, book p69 (lymphatic row) and p70 (immune row). SLO 4 and SLO 6. **RE-TEST** of exam-16 Q12, where she chose lymphatic instead of immune.

6. Structure at a different level of organization: (a) neuron 1

Kidney, liver and lung are all **organs**. A neuron is a single specialized **cell**, so it sits one level lower in the pattern of increasing complexity.

Source: Ch 5, section 5.1 levels of organization, cellular and organ levels (book p65 to p66), and Exercise Section I, MCQ 14, book p76. SLO 2. **NEW**

7. The disaccharide also called the transport sugar: (d) sucrose 1

Sucrose is glucose plus fructose. It is the sugar plants move around, and it is also cane sugar, the most familiar disaccharide. Maltose is malt sugar, lactose is milk sugar, and fructose is not a disaccharide at all.

Source: Ch 6, section 6.6.2 Disaccharides (book p85 to p86), and Exercise Section I, MCQ 9, book p92. SLO 6. **NEW**

8. All RNA nucleotides differ from DNA nucleotides in having a different: (c) pentose sugar 1

In RNA the pentose sugar is **ribose**; in DNA it is **deoxyribose**. That difference is in **every single** nucleotide. The nitrogen base differs only in one case, uracil in place of thymine, so it is not true of all of them. The phosphate group is the same, and there is no carboxylic group in a nucleotide.

Source: Ch 6, section 6.9 Structure of RNA, book p89, and Exercise Section I, MCQ 7, book p92. SLO 9. **NEW**

9. The two bonds shown by wavy lines in ATP: (a) high energy bonds 1

The book: "The covalent bonds between phosphate groups are called high energy or energy rich bonds. The two bonds linking the phosphate group together are represented by a specific sign of wavy lines." Hydrogen bonds hold the two DNA strands together, a peptide bond joins two amino acids, and double bonds occur in unsaturated fatty acids.

Source: Ch 7, section 7.6.2 and Fig. 7.8, book p101. SLO 6. **RE-TEST** of exam-16 Q17, where she chose hydrogen bonds.

10. Products of the light reactions: (b) ATP, NADPH and oxygen 1

In the light dependent reactions the electron transport chain drives **ATP** synthesis, photosystem I reduces NADP^+ to **NADPH**, and photolysis of water releases **oxygen**, which leaves the leaf through the stomata. NADP and PGA are not products of this phase. 3PGA belongs to the Calvin cycle.

Source: Ch 7, sections 7.7.1 and 7.7.1.1, book p102 to p104, and Exercise Section I, MCQ 14, book p110. SLO 7. **NEW**

Answer letters: b, c, a, d, b, a, d, c, a, b (3 A, 3 B, 2 C, 2 D; no letter repeats twice in a row).

SECTION B: ANSWER KEY (6 × 4 = 24 MARKS)

Q.2 (Chapter 1) The spread of malaria 4

(i) Hypothesis (1): “Mosquitoes transmit Plasmodium”, that is, malaria is carried from a sick person to a healthy person by mosquito bite.

(ii) Ronald Ross’s sparrow experiment (2): Ross allowed female mosquitoes to bite **sparrows that were suffering from malaria** (1). After a few days the same mosquitoes were allowed to bite **healthy sparrows**, and those healthy sparrows became ill with malaria (1). *Also accept the first Ross experiment:* he allowed a female Anopheles to bite a malarial patient, dissected it, and found Plasmodium multiplying in the mosquito’s stomach.

(iii) Test on human beings (1): An **Italian biologist** let an Anopheles mosquito bite a malarial patient, kept the mosquito for a few days, and then allowed it to bite a healthy man. That man later developed malaria, which confirmed the hypothesis.

Source: Ch 1, section 1.8 (Spread of malaria) and Fig. 1.5, book p15 to p16. SLO 8. **NEW STEMS**, topic previously covered in exam-01, 07, 08 and 11.

Q.3 (Chapter 2) Differences 4

(i) Bacteria and Protists (2), any two, 1 mark each:

Bacteria	Protists
Prokaryotic cells (no membrane bound nucleus)	Eukaryotic cells (true nucleus present)
Their cell walls contain peptidoglycan	If a wall is present it has no peptidoglycan
They form their own domain, Bacteria	They are a kingdom inside domain Eukarya
Always unicellular	Unicellular or colonial, mostly aquatic

(ii) Fungi and Plants (2), any two, 1 mark each:

Fungi	Plants
Cell wall made of chitin	Cell wall made of cellulose
No chlorophyll and no chloroplasts	Chlorophyll present inside chloroplasts
Saprotrophic, absorptive nutrition from dead matter	Autotrophic, they make their own food by photosynthesis
Examples: bread mould, yeast, mushroom	Examples: moss, mustard

Source: Ch 2, section 2.4 (Domain Bacteria; four kingdoms of Domain Eukarya) and modes of nutrition, book p24 to p26, and Exercise Section II, Q5(a) and Q5(b), book p29. SLO 3 and SLO 4. **NEW**

Q.4 (Chapter 3) SER and the salivary gland cell 4

(i) A cell rich in SER (2), any two, 1 mark each:

- It will synthesize **lipids, including steroids**, more efficiently.
- It will **detoxify harmful substances** more efficiently.
- If it is a muscle cell, it will carry out **contraction** more efficiently, because SER has an important role in that process.
- *Also accept:* better mechanical support and better transport of substances within the cell, which the book gives for the ER as a whole.

(ii) Organelles abundant in a salivary gland cell (2), any two named with a reason, 1 mark each:

- **Ribosomes and rough endoplasmic reticulum:** a gland cell makes large amounts of protein (the enzyme amylase), and protein synthesis happens on ribosomes attached to the RER.
- **Golgi apparatus:** it stores, modifies and packages the material and carries out cell secretion, which is exactly what a gland does.
- **Mitochondria:** making and secreting so much material needs a lot of ATP, and mitochondria are the power house of the cell.

Source: Ch 3, Endoplasmic reticulum and Ribosomes (book p34), Golgi apparatus and Mitochondria (book p35 to p36); the questions themselves are Exercise Section III Q3 and Exercise Section II Q11, book p46 to p47. SLO 3 and SLO 7. **NEW**

Q.5 (Chapter 5) Plant tissues 4**(i) Xylem and phloem (2), 1 mark each side:**

Xylem	Phloem
Long dead cells (xylem vessels)	Living phloem cells
Transports water from the root up to the leaves	Carries the prepared food from the leaf to the other parts of the plant

(ii) Dermal tissue and ground tissue (2), 1 mark each side:

Dermal tissue	Ground tissue
Forms the outer covering of the plant (the epidermis)	Lies inside, between the dermal and the vascular tissue
Covers and protects the plant and checks water loss	Site of photosynthesis , and it also gives support and stores food

Source: Ch 5, section 5.3 Plant tissues (book p70) and section 5.3.2 structure of a leaf (book p72); the pairs are Exercise Section II, Q1(c) and Q1(f), book p76. SLO 7 and SLO 8. **NEW** (xylem and phloem had appeared only once, as a single MCQ).

Q.6 (Chapter 6) Carbohydrates and chromatin 4

(i) Hydrates of carbon (2): Carbohydrates are compounds of carbon, hydrogen and oxygen in the ratio **1 : 2 : 1** (1 mark). Because the hydrogen and oxygen are present in the same proportion as in water, they are literally called hydrates of carbon, and their general formula is **$C_x(H_2O)_y$** (1 mark). For glucose this gives $C_6(H_2O)_6$, that is $C_6H_{12}O_6$.

(ii) Chromatin (2): Chromatin is composed of **DNA together with proteins called histones** (1 mark). The coiling of DNA around the histones forms bead like structures called **nucleosomes** (1 mark).

Source: Ch 6, section 6.5.1 Structure of carbohydrates (book p84 to p85) and section 6.8 with Fig. 6.14 (book p89); the wording is taken from Exercise Section II Q5 and Section III Q2, book p92. SLO 5, SLO 6 and SLO 8. **NEW**

Q.7 (Chapter 7) Respiration 4

(i) Glycolysis (2): Glycolysis takes place in the **cytoplasm** of the cell, and it is common to both aerobic and anaerobic respiration (1 mark). One molecule of glucose is oxidized into **two molecules of pyruvic acid** with a yield of **two ATP** molecules (1 mark). The book prints it as:



Also accept the fuller form from section 7.8.3, which adds 2 NADH.

(ii) The four stages of aerobic respiration (2): glycolysis, formation of acetyl CoA, the Krebs cycle and the electron transport chain, in that order (1.5 marks for all four in the right order). Oxygen is used in the **electron transport chain**, where it is the final electron acceptor and combines with hydrogen ions to form water (0.5 mark).

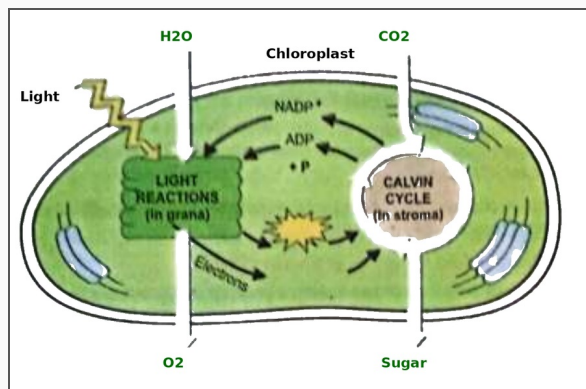
Source: Ch 7, section 7.8.1 (book p105), section 7.8.3 (book p106 to p107) and Key Points 13 and 16, book p108. SLO 8. Topic last touched in exam-02 (April); stems rewritten.

SECTION C: ANSWER KEY (2 × 8 = 16 MARKS)

Q.8 (Chapter 7) Photosynthesis 8

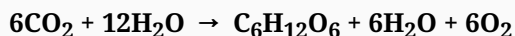
(i) The four labels (4), 1 mark each:

Label	Answer	Why
A	Water, H₂O	It enters the chloroplast at the top and feeds the light reactions, where it is split by photolysis.
B	Carbon dioxide, CO₂	It enters at the top on the right and feeds the Calvin cycle in the stroma.
C	Oxygen, O₂	It leaves from the light reactions side and is released out of the leaf through the stomata.
D	Sugar (glucose)	It leaves from the Calvin cycle side and is the food the plant has made.



The same figure with the four labels filled in

(ii) The balanced equation (2):



1.5 marks for the equation written and balanced exactly as the book prints it. 0.5 mark for naming **Light** above the arrow and **Chlorophyll** below it.

Marking note: watch the coefficients. Writing $6\text{C}_6\text{H}_{12}\text{O}_6$ instead of $\text{C}_6\text{H}_{12}\text{O}_6$ costs 0.5 mark, which is the exact slip made in exam-17 Q.2(i)(a). Also accept the shorter Chapter 6 form, $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$, book p83.

(iii) The two energy rich molecules (2): **ATP** (1 mark) and **NADPH** (1 mark). The book: “Once the light reactions produce ATP and NADPH, a photosynthetic cell can fix carbon dioxide to synthesize sugar molecules.”

Source: Ch 7, section 7.7 with the printed equation (book p102), Fig. 7.9 overview of photosynthesis (book p103), section 7.7.1.1 step 3 (book p104) and section 7.7.1.2 (book p104). SLO 7. **NEW** figure, never used in any earlier paper. Part (ii) is a **RE-TEST**.

Q.9 (Chapter 4) Meiosis 8**(i) Four differences between meiosis I and meiosis II (4), 1 mark each, any four:**

Meiosis I	Meiosis II
It is the reduction division: a diploid cell gives haploid cells	The chromosome number is not reduced: haploid stays haploid
In prophase I homologous chromosomes pair up by synapsis and form bivalents	In prophase II there is no pairing and no bivalent
Crossing over takes place at the chiasmata in prophase I	No crossing over takes place
In anaphase I the whole homologous chromosomes separate and move to opposite poles	In anaphase II the centromeres divide and the sister chromatids separate
It follows interphase I, in which the DNA is replicated	It follows interphase II, in which there is no further DNA replication
It produces two cells	It brings the total up to four cells
Prophase I is the longest phase of meiosis	Prophase II is short

(ii) The 18 chromosome cell (3), 1 mark each:

- **(a)** At the end of meiosis I each cell has **9 chromosomes**, because meiosis I halves the number.
- **(b)** At the end of meiosis II each cell still has **9 chromosomes**, because meiosis II separates the sister chromatids and does not reduce the number again.
- **(c)** A total of **four** daughter cells is formed.

(iii) Why the daughter cells are not identical (1): because **crossing over** in prophase I exchanges genetic material between the non-sister chromatids of the homologous pair, so each chromosome now carries a new combination of genes. *Also accept:* the independent combination of the maternal and paternal chromosomes as they go to the poles in anaphase I.

*Source: Ch 4, section 4.5 Meiosis, Prophase I to Cytokinesis II (book p53 to p57), section 4.4.1 Significance of meiosis (book p57 to p58); the difference table follows Exercise Section II Q6, book p62. SLO 2, SLO 3 and SLO 4. **NEW** for part (i); part (i) is also the **RE-TEST** of exam-17 Q.3(ii)(c), where she compared the wrong two things in a difference table.*

Syllabus check. Every question on this paper was traced to a printed section, figure or exercise item in Chapters 1 to 7 of her own textbook, and to a stated SLO from the list on page 1 of that chapter. **No question, no option and no part of this key refers to Chapter 8 (Plant Physiology), Chapter 9 (Plants Reproduction) or Chapter 10 (Evolution).**